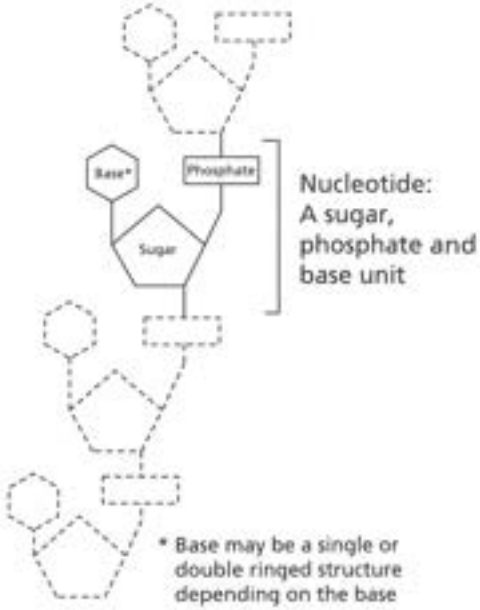
18 Role and structure of chromosomes

Chromosomes are housed in the cell's nucleus. Chromosomes contain a nucleic acid called DNA, or deoxyribonucleic acid *(deoxy* means 'less oxygen'). DNA carries an organism's genetic information*. The many tiny units that make up the DNA molecule are genes. A gene is a series of nucleotides and contains specific information about which characteristics the organism will inherit.

* DNA also has assorted proteins attached to it.

The gene is the smallest unit of heredity and genes are linked by a sugar-phosphate 'backbone'. The name given to the unit that is formed by the deoxyribose, the phosphate and a nitrogen base, is *nucleotide*. Weak attractions ('hydrogen bonding') between strands cause them to twist into a double helix shape (think of a spiral staircase). The whole unit is called a chromosome.

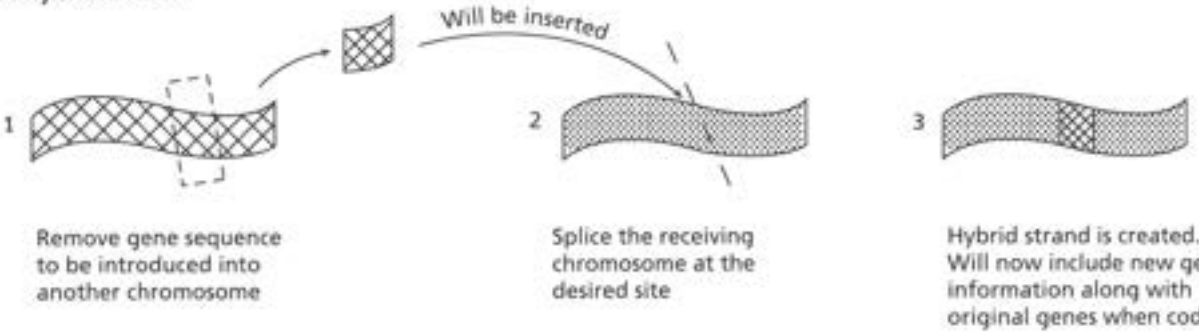


1 *Drosophilla melanogaster,* the common fruit fly, has four chromosomes. How many did it receive from its mother?

Genetic manipulation: How humans can interfere with genetic material

Genetic manipulation has been occurring for some time now and, despite protests, it will continue. Many techniques used for genetic manipulation depend on the DNA's ability to repair itself (to a degree) should it become slightly damaged in the process.

DNA hybridisation



Technically, a hybrid is a cross between two genetically different organisms. Hybrid DNA is created by breaking (denaturing) the protein bonds on the strand without completely destroying them. (Consider them loosened.) Heat is used to denature proteins and a slight elevation in temperature can cause bonds to loosen and break but, if handled carefully, they can reform. In this way unrelated pieces of genetic material can be joined to a strand of DNA. This technique can be used to create some forms of genetically modified organisms (GMOs).

2 In the USA, about 39% of crops contain some genetically modified plants, whereas in China it is 2% and India it is 0.1%. Discuss why this might be the case.

3 Define *nucleotide.*

4 What causes the two strands in DNA to form the double helix shape?

5 Define *denatured,* with respect to DNA.

What do the following terms define?

6 *Nucleic acid located in nucleus of cell, which is the basis of heredity.*

7 *The deliberate alteration of the DNA inside a cell's nucleus with the aim of modifying that organism.*

8 *Producing a genetically identical duplicate of an organism.*

9 *Cell body (made up of genes) that plays a large role in inheritance.*

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